

A Three-Isomorphism From the Prochazka–Sanchez Free Space to ℓ_1

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Status

This is a partial result for the metric space M introduced by Prochazka and Sanchez-Gonzalez in [1]. It improves the upper bound recorded in the source paper for the Banach–Mazur distance between $\mathcal{F}(M)$ and ℓ_1 , but it does not determine the exact constant.

Source Question

Let

$$M = \{\emptyset\} \cup \mathbb{N} \cup F, \quad F = \{A \subset \mathbb{N} : 1 \leq |A| < \infty\},$$

with graph edges $\emptyset \sim n$ for $n \in \mathbb{N}$, and $n \sim A$ whenever $n \in A$. The corresponding shortest-path metric satisfies

$$\begin{aligned} d(\emptyset, n) &= 1, & d(\emptyset, A) &= 2, & d(n, m) &= 2, \\ d(n, A) &= 1 \text{ if } n \in A, & d(n, A) &= 3 \text{ if } n \notin A, \end{aligned}$$

and

$$d(A, B) = 2 \text{ if } A \cap B \neq \emptyset, \quad d(A, B) = 4 \text{ if } A \cap B = \emptyset.$$

The source paper proves that M does not embed into any $L_1(\mu)$ with distortion strictly less than 2, but that $\mathcal{F}(M)$ is isomorphic to ℓ_1 . It then asks whether

$$d_{BM}(\mathcal{F}(M), \ell_1) = 2.$$

The paper’s argument gives only $d_{BM}(\mathcal{F}(M), \ell_1) \leq 4$, while the L_1 -distortion obstruction gives $d_{BM}(\mathcal{F}(M), \ell_1) \geq 2$.

Proof Intuition

The usual proof that a bounded uniformly discrete countable metric space has free space isomorphic to ℓ_1 sends every non-basepoint Dirac mass to an unweighted unit vector. For this particular M , that loses a factor 2 on the points $A \in F$, because $\|\delta_A\|_{\mathcal{F}(M)} = d(A, \emptyset) = 2$. Rescaling just those coordinates by 2 removes that loss. The only remaining expensive pairs are incident pairs $n \in A$, where the image distance is $\|e_n - 2e_A\|_1 = 3$ although $d(n, A) = 1$.

Theorem 1. *For the metric space M above,*

$$2 \leq d_{BM}(\mathcal{F}(M), \ell_1) \leq 3.$$

Proof. The lower bound is due to [1]. Indeed, if $S : \mathcal{F}(M) \rightarrow \ell_1$ were an isomorphism with $\|S\| \|S^{-1}\| < 2$, then $x \mapsto S\delta_x$ would be a bi-Lipschitz embedding of M into ℓ_1 with distortion strictly less than 2. This contradicts Corollary 3 of [1], which says that M does not embed into any $L_1(\mu)$ with distortion < 2 .

For the upper bound, let $E = \mathbb{N} \sqcup F$, and identify $\ell_1(E)$ with the ℓ_1 -sum of the singleton coordinates and the finite-subset coordinates. Define $f : M \rightarrow \ell_1(E)$ by

$$f(\emptyset) = 0, \quad f(n) = e_n \quad (n \in \mathbb{N}), \quad f(A) = 2e_A \quad (A \in F).$$

The universal property of the Lipschitz-free space gives a unique bounded linear operator $T : \mathcal{F}(M) \rightarrow \ell_1(E)$ with $T\delta_x = f(x)$, and $\|T\| = \text{Lip}(f)$.

We compute the Lipschitz constant from the displayed distance table. For pairs involving $\emptyset$,

$$\frac{\|f(n)\|_1}{d(\emptyset, n)} = 1, \quad \frac{\|f(A)\|_1}{d(\emptyset, A)} = \frac{2}{2} = 1.$$

For two distinct singleton points,

$$\frac{\|e_n - e_m\|_1}{d(n, m)} = \frac{2}{2} = 1.$$

For $n \in \mathbb{N}$ and $A \in F$,

$$\|f(n) - f(A)\|_1 = \|e_n - 2e_A\|_1 = 3.$$

Thus the quotient by $d(n, A)$ is 3 when $n \in A$, and 1 when $n \notin A$. Finally, for $A \neq B$ in F ,

$$\|f(A) - f(B)\|_1 = \|2e_A - 2e_B\|_1 = 4,$$

so the quotient is 2 if $A \cap B \neq \emptyset$ and 1 if $A \cap B = \emptyset$. Hence $\|T\| \leq 3$.

The map is onto $\ell_1(E)$. Its inverse on the canonical basis is

$$T^{-1}e_n = \delta_n, \quad T^{-1}e_A = \frac{1}{2}\delta_A.$$

Since $\|\delta_n\|_{\mathcal{F}(M)} = 1$ and $\|\delta_A\|_{\mathcal{F}(M)} = 2$, the operator $T^{-1} : \ell_1(E) \rightarrow \mathcal{F}(M)$ has norm at most 1, because the unit ball of $\ell_1(E)$ is the closed convex hull of the signed coordinate vectors. Therefore

$$\|T\| \|T^{-1}\| \leq 3,$$

which proves the upper bound. □

Verification Notes And Failed Full-Closure Routes

The diagonal family $T_{s,t}\delta_n = s\delta_n$, $T_{s,t}\delta_A = t\delta_A$ cannot improve the constant 3. Its distortion is

$$\max\{s + t, t, s\} \cdot \max\{s^{-1}, 2t^{-1}\},$$

and the minimum over $s, t > 0$ is attained at $t = 2s$, giving 3. Thus any proof of the conjectural value 2, if true, must use a substantially non-diagonal representation.

Finite L_1 -cut LPs were run for the truncations

$$M_n = \{\emptyset\} \cup \{1, \dots, n\} \cup \{A \subset \{1, \dots, n\} : A \neq \emptyset\}.$$

They gave optimal L_1 -metric distortions 1 for $n = 2$, $4/3$ for $n = 3$, and approximately 1.4615384615 for $n = 4$ using a symmetry-reduced cut cone. These computations are consistent with the source paper's asymptotic lower bound 2, but they do not prove a Banach–Mazur lower bound above 2 for $\mathcal{F}(M)$.

Novelty Check

Local run indexes were searched for arXiv:1311.4584, the title, and the phrases “Banach–Mazur distance”, “Prochazka Sanchez-Gonzalez”, “Lipschitz-free”, and “ ℓ_1 ”. No exact prior packet was found. Web searches on 2026-06-20 for the exact source question and nearby phrases found the source paper [1], the Dalet–Kaufmann–Prochazka isometric characterization for tree metrics [2], and the Aliaga–Petitjean–Prochazka embedding theorem for free spaces over subsets of real trees [3]. Those later tree results do not directly apply to the present graph metric, which is not a tree metric.

Human Review Recommendation

The proof of the upper bound is elementary and should be quick to verify. The packet should be reviewed as a genuine but modest partial improvement, not as a solution of the exact-distance problem.

References

- [1] A. Prochazka and L. Sanchez-Gonzalez, *Low distortion embeddings into Asplund Banach spaces*, arXiv:1311.4584.
- [2] A. Dalet, P. L. Kaufmann, and A. Prochazka, *Characterization of metric spaces whose free space is isometric to ℓ_1* , arXiv:1502.02719.
- [3] R. J. Aliaga, C. Petitjean, and A. Prochazka, *Embeddings of Lipschitz-free spaces into ℓ_1* , arXiv:1909.05285.